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System-of-systems effectiveness evaluation of UAV swarm: A data-driven perspective



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Abstract System-of-Systems Effectiveness Evaluation (SEE) is crucial for systematically assessing Unmanned Aerial Vehicle Swarms (UAVS). However, due to the strong nonlinear, hierarchical, and emergent characteristics of UAVS, traditional SEE methods based on experience and mathematical analysis are challenging to apply to such complex systems. In this article, we propose a Data-driven SEE framework that leverages deep learning to establish the relationship between system-of-systems indicators and effectiveness, effectively capturing the complex characteristics of UAVS. The system-of-systems indicators are first transformed by two-dimensional loop stacking and then processed by a residual neural network-based evaluation network. Furthermore, to address the practical need to mitigate overestimation in effectiveness evaluation, a skewed loss function is designed to adjust overestimation during the training process. Since training a Data-driven model requires a large number of labeled samples, which are obtained through time-consuming simulation-based annotation, we develop an active learning strategy based on cluster iteration and random greedy sampling. This strategy selects a limited number of high-value samples for simulation labeling while maintaining model accuracy. The experimental results demonstrate that this model can quickly, accurately, and efficiently evaluate the effectiveness of unmanned swarm systems. Additionally, the data-driven SEE framework shows strong potential for UAVS system analysis and optimization.

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1. Introduction

With the rapid progress in unmanned system techniques and swarm intelligence, Unmanned Aerial Vehicle Swarm (UAVS) has gained widespread applications in industrial, emergency problems, and military.^{1–3} System-of-Systems (SoS) effectiveness refers to the degree to which a system can achieve its intended goals under complex conditions during specific tasks,⁴ making it a crucial factor in the development of

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unmanned swarm systems. As an indispensable link in demonstrating UAVS capabilities, system-of-Systems Effectiveness Evaluation (SEE) has received increasing attention.

The aim of SEE for UAVS is to evaluate (predict) the effectiveness UAVS systems can achieve in specific tasks, which is accomplished by establishing an evaluation framework to assess their effectiveness based on SoS indicators before executing tasks. Through providing a comprehensive, precise, and real-time evaluation, SEE can provide critical information and thus guide the decision process in subsequent missions. The SEE of UAVS is a complex, multi-dimensional problem due to the UAVS SoS's nonlinearity, hierarchy, and emergent properties.⁵ The vast swarm sizes, intricate interconnections, and the large number of diverse indicators further complicate the SEE. Given these challenges, this paper aims to establish a holistic evaluation framework to accurately assess the effectiveness of UAVS.

Over the past decades, various SEE methodologies have been proposed, which can be categorized into four main groups: experience-driven, mathematical analysis-driven, simulation-driven, and Data-driven SEE. The experience-driven method⁶ and mathematical analysis-driven approach aim to establish evaluation models from SoS indicators to effectiveness based on expert knowledge and analytical mathematical functions. Due to the complexity and emergence of UAVS systems, the difficulty of constructing a credible method through experts' subjective experience⁷ or mathematical models to evaluate effectiveness is gradually increasing. For such complex systems, simulation-based SEE methods are increasingly applied. The operation and interaction of UAVS in actual environments can be simulated, thereby obtaining more objective and accurate evaluation results.⁸ However, the acquisition of effectiveness for each task requires time-consuming simulations and statistics, which makes it difficult to provide rapid results for a specific system.

Utilizing simulated historical or practical data, the Data-driven approach can establish an intelligent evaluation model that deeply reflects the complexity of swarm systems and achieves precise evaluation. This method not only avoids human interference, improving the objectivity and accuracy of evaluation results, but also enables optimization of the parameters, strategies, or structures of the swarm system.⁹ As an advanced technique within the Data-driven field, deep learning-based evaluation¹⁰ can learn from extensive historical data, continuously improving the model's generalization ability, and thus providing high-precision evaluation in situations with abundant evaluation data.

The deep learning-driven SEE still faces several challenges related to model-demand alignment and data dependencies. First, deep learning struggles to directly align the established UAVS model with real-world application requirements. On one hand, the extracted indicator systems typically consist of one-dimensional, non-sequential features, which are inconsistent with the typical feature format used in advanced deep learning. On the other hand, general prediction models rarely consider the impact of bias,¹¹ which is the discrepancy between predicted and actual effectiveness values. In application scenarios, when the model's predicted effectiveness exceeds the actual effectiveness, this can lead to an overestimation of the system's capabilities. If this configuration is used to perform a task, the UAVS system might fail because its actual effectiveness does not meet expectations. Therefore, the impact of pre-

diction biases must be accounted for in real-world applications. Secondly, regarding data dependencies, training Data-driven evaluation models requires a large amount of extensive training data. However, obtaining effectiveness data for UAVS requires running system simulations for annotation, which is time-consuming and relatively expensive.¹²

The overall goal of this paper is to develop a Data-driven framework that addresses the aforementioned challenges of model-demand alignment and data dependencies. To achieve this goal, this paper proposes a Data-driven SEE framework, including the main evaluation network: a residual-based evaluation network with skewed loss, and an active learning strategy to select high-value samples for annotation and model training. The main contributions of this paper can be summarized as follows:

- (1) A comprehensive Data-driven SEE evaluation framework is proposed, which integrates a deep learning-based evaluation network: Residual Network with Skew loss (ResNet-Skew) with an efficient active learning-based sample selection strategy. This framework enables fast, accurate evaluation of UAVS system effectiveness with reduced labeling costs, supports system configuration optimization, and demonstrates strong generalization ability through transfer learning across different operational scenarios.
- (2) A ResNet-Skew evaluation network is designed for fast and accurate Data-driven evaluation of UAVS. This evaluation model integrates the ResNet model to capture feature dependencies in SoS indicators processed by loop stacking methods. A skewed loss function is also applied to reflect the asymmetric risk preference in UAVS operations, penalizing overestimation more heavily to meet mission-critical safety requirements.
- (3) An active learning strategy is proposed to select high-value samples for labeling through the swarm simulation system. By combining the cluster-based method and random greedy sampling, the algorithm can reduce the required number of labeled samples significantly while maintaining evaluation accuracy.
- (4) The proposed framework is extensively validated through comparisons with state-of-the-art baselines, ablation and utility analysis, as well as generalization and sensitivity tests, demonstrating its robustness and practical applicability across various scenarios.

The rest of this paper is organized as follows. [Section 2](#) reviews the related work on SEE. The framework of the proposed Data-driven SEE is described in [Section 3](#). [Sections 4 and 5](#) elaborate on the deep learning-based SEE and active learning strategy of the framework. [Section 6](#) presents the experiment of the proposed method. Finally, [Section 7](#) concludes this paper.

2. Related works

2.1. SEE method

A variety of approaches have been proposed to solve SEE problems. The evaluation process typically begins by defining a comprehensive set of SoS indicators, followed by the

calculation of effectiveness based on these indicators. For the effectiveness calculation, the experience-driven and mathematical analysis-driven methods are first applied. The experience-driven method, such as the Analytic Hierarchy Process (AHP),¹³ relies on expert judgment or empirical data to assign weights to indicators in SEE. AHP is often combined with techniques like fuzzy theory¹⁴ and TOPSIS¹⁵ to better represent expert knowledge. Li et al.¹⁶ combined AHP with the entropy weight method to form a multi-index evaluation method for evaluating the coverage effectiveness of a remote sensing satellite. Furthermore, He et al.¹⁷ adopted a multidimensional normal cloud model to determine SoS indicators' weights and thus evaluated the Communication. However, the application of these effectiveness evaluation methods is mainly based on expert experience, thereby always tending to be highly subjective and resulting in evaluation outcomes that are hard to trust.

Mathematical analysis-driven SEE methods construct models to quantify the relationships between SoS indicators and effectiveness. These models, like ADC¹⁸ and the entropy weight method,¹⁹ provide a precise way to compute system performance by analyzing how changes in indicators affect outcomes. Owing to the complexity and uncertainty of swarm systems, traditional mathematical analysis often simplifies the system, which causes a reduction in the accuracy and reliability of the results. In a related research area, resilience evaluation of UAVS, complex network modeling approaches are widely applied, where the swarm is represented as a graph, and indicators such as connectivity, robustness, and efficiency are mathematically analyzed.²⁰ In terms of effectiveness, it can be represented in a complex network. Hu et al.²¹ calculated the communication effectiveness based on the path lengths of networks.

In contrast, simulation-driven SEE methods leverage simulation models to evaluate SoS effectiveness. These methods simulate the behavior of the system under specific conditions defined by the SoS indicators. By executing these simulations, effectiveness can be determined by the results of the system's performance in a virtual environment.²² Jia et al.⁸ proposed an effectiveness evaluation method for UAVS systems through the established simulation system based on the system dynamics model. Furthermore, Agent-Based Modeling (ABM)²³ is frequently employed as a simulation-driven evaluation technique, especially for UAVS. Quan et al.²⁴ and Jia et al.²⁵ employed ABM to establish the simulation system of the UAVS in tracking tasks and thus applied it in evaluating and further optimizing its effectiveness. Furthermore, Kong et al.²⁶ and Bai et al.²⁷ respectively established multi-agent simulation systems to evaluate the effectiveness and resilience of UAVS under resource supplementation and communication limits. Simulation-driven evaluation methods yield results with high credibility; however, rapid assessment of equipment systems is hindered by high modeling costs and lengthy simulation modeling and execution times.

2.2. Data-driven SEE

The large volumes of historical data generated during large-scale simulations and actual task executions provide valuable opportunities to uncover complex relationships between system entities. Data-driven SEE methods utilize machine learn-

ing techniques to model the non-linear relationships between input indicators and output effectiveness, enabling the performance evaluation by learning directly from historical data. These methods establish a complex link between SoS indicators and effectiveness, facilitating accurate predictions and assessments.

Due to their ability to rapidly and precisely assess system performance, Data-driven SEE methods have garnered substantial research interest. Traditional machine learning approaches, such as the BP Neural Network (BPNN),⁷ Relevance Vector Machine (RVM),²⁸ and Support Vector Machine (SVM),²⁹ have been applied to successfully evaluate systems with simpler structures. These techniques rely on straightforward models to establish a mapping relationship between SoS indicators and corresponding effectiveness. Yu et al.³⁰ developed some basic neural networks to evaluate the effectiveness of UAVS. To enhance the performance of the Data-driven model, they adopt feedback-based attribute extraction and anti-causality detection to select more valuable indicators and training data. Zhao et al.³¹ adopted the Deep Belief Network (DBN) to evaluate the effectiveness of the anti-missile equipment system. In this Data-driven model, the parameter of the DBN is optimized by the grey wolf optimization algorithm. Han et al.⁷ combined data from a simulation and AHP methods to train a BPNN for effectiveness evaluation. Zhao et al.²⁹ employed SVM to assess the effectiveness of different disposition strategies. However, as system complexity grows and the volume of simulation and real-world data increases, traditional small-data models and machine learning techniques are becoming less effective. A promising solution to this challenge is introducing deep learning techniques, which have the potential to reconstruct simulation processes and enhance effectiveness evaluation.³² Deep Neural Networks (DNN),³³ as foundational deep learning models, have already been applied to SEE problems. Additionally, autoencoders,³⁴ known for their robust dimensionality reduction capabilities, can effectively represent the information contained in high-dimensional indicators with fewer dimensions, making them a valuable tool for SEE applications. Qi et al.¹⁰ were among the first to integrate sparse representations into autoencoders, allowing for the evaluation process and enabling effective assessment of high-dimensional effectiveness.

Convolutional Neural Network (CNN), widely regarded as one of the most powerful deep learning methods, is particularly effective in extracting complex, deep features across multi-dimensional data. Through convolutional structures, CNN can explore and capture the relationships between different SoS indicators. Li et al.³⁵ employed 1D-CNN to process one-dimensional SoS indicators in SEE problems. Zhao et al.³⁶ applied a CNN with Bayesian optimization to learn the relation between the 1D indicators and the effectiveness of anti-missile equipment based on the simulation data. Moreover, by transforming SoS indicators into two or more dimensions, CNN can more effectively uncover implicit relationships between indicators, improving the speed and accuracy of the evaluation process. Besides CNN, advanced deep learning models such as Graph Neural Networks (GNNs), which can express the complex relation of complex point and edges,³⁷ have also shown success in swarm coordination and communication optimization,^{38,39} although their direct application to SoS effectiveness evaluation remains an open and under-

explored area; such models can offer a potential direction for future research.

While current research in Data-driven SEE has largely focused on traditional machine learning approaches, deep learning-based methods, especially CNN-based networks, offer significant advantages in extracting complex relationships from SoS indicators. These methods are particularly well-suited for capturing the emergent behaviors and nonlinearities present in UAVS systems. Thus, the application of CNN to SEE remains a promising area for further exploration. Among much of the existing research on Data-driven SEE, they have not sufficiently considered the challenges related to data collection, particularly the high costs and complexities of gathering large-scale, high-quality data from simulation systems or the real world.⁴⁰ Moreover, these approaches often fail to consider the alignment between the trained models and the actual requirements of the systems.¹¹ In real-world applications, ensuring that model predictions align with the specific demands of UAVS is critical to achieving accurate and reliable effectiveness evaluations.

To address these gaps, this study will explore methods for overcoming the challenges of data collection and propose solutions to ensure that the trained models align with real-world application requirements. The focus will be on developing a deep learning-based SEE framework that not only captures the complex relationships among SoS indicators but also requires a limited cost to be established.

3. Effectiveness evaluation framework of UAVS

3.1. SEE problem for UAVS

SEE assesses the target's effectiveness based on predefined SoS indicators, helping to identify gaps and optimize the effectiveness under varying conditions. The evaluation of UAVS systems' effectiveness is a critical component in the development of unmanned system configurations. Given its high accuracy and low time cost, Data-driven SEE is particularly well-suited for assessing UAVS. By leveraging historical simulation data to construct an evaluation model, Data-driven SEE can learn the relationship between swarm SoS indicators and overall SoS effectiveness.

Mathematically, Data-driven SEE for UAVS can be defined as follows: Based on the task performance data collected from the UAVS simulation system, a simulated dataset $O_s = \{O_{s1}, O_{s2}, \dots, O_{sm}\}$ can be established, consisting of multiple UAVS SoS indicators and corresponding effectiveness data. The SEE data for UAVS i , denoted as $O_{si} = \{D_i, E_i\}$, consists of $D_i = [D_{i1}, D_{i2}, \dots, D_{im}]$, which represents m SoS indicators, and E_i , which denotes the simulated SoS effectiveness. The goal of the Data-driven SEE for UAVS is to construct an evaluation model capable of predicting the effectiveness E_i through the given SoS indicators D_i . Thus, the model can be expressed as $E_i = f(D_i)$, where $f(\cdot)$ is the prediction function, the core objective of the SoS effectiveness model. Once the model is fully trained, new UAVS, along with their respective SoS indicators, can be evaluated and analyzed.

3.2. UAVS simulation and SoS indicators

The simulation system serves as the medium for assessing the effectiveness of unmanned swarms. To provide a carrier for the evaluation, a simulation system for UAVS is established for further effectiveness evaluation. It is worth noting that our Data-driven SEE framework is not limited to this simulation system, but can adapt to different missions. Considering that UAVS are widely applied to target detection,³ a typical scenario for collaborative detection and tracking tasks based on Refs. 19, 24 and 25 is designed and simulated. The designation details of the simulation system are described in Appendix A. The mission requires detection drones and communication drones to detect and track multiple dynamic targets (ground vehicles in this task) in a multi-obstacle urban environment. The UAVS will search the map for the vehicles and track targets once located. If one target is tracked for a certain period, the tracking task for that target is considered successful. In order to increase the complexity of the task, the target vehicles are considered to cause drones to become disabled during the tracking process. Fig. 1 shows some scenarios of the simulated missions, including the 3D scenario, 2D scenario, and a local view of a group of UAVS, including 4 detection UAVs and 1 communication UAV is tracking one blue vehicle target. In these scenarios, the UAVS and target vehicles have been amplified for better display. The red UAV represents the communication UAV, and the yellow UAV denotes the UAV for detection. The blue vehicle represents target vehicles tracked by UAVS.

Utilizing the established scenario and simulation system, the key indicators that influence the effectiveness of UAVS and the method to evaluate the SoS effectiveness are defined. As shown in Table 1, UAVS are categorized into communication UAVs and detection UAVs. Based on their characteristics, key SoS indicators include quantity, maximum speed, acceleration, and detection range (or communication range for communication UAVS).

These indicators are chosen because they can be adjusted before the mission, allowing for better mission adaptation and improved performance. Indicators that cannot be modified during the task are excluded from the evaluation, as they fall outside of control. To enhance the universality and scalability of the SEE model, the relevant indicators of targets to be tracked, similar to UAVS SoS indicators, are also included in the model's input. The inclusion of targets' indicators is crucial, as it introduces uncertainty into the mission. This prevents the model from relying on fixed parameters and ensures adaptability to the dynamic, unpredictable nature of real-world scenarios. Therefore, 12 main factors are considered, which can significantly influence UAVS' effectiveness.

In terms of the measurement of SoS effectiveness, the task completion rate is the primary factor based on the characteristics of the task. This indicator represents the proportion of targets detected and tracked within the specified time in the task. Assuming that the total number of target vehicles is N_t and the number of targets detected and tracked by the drone swarm is N_d , the completion rate of the task M_c is

$$M_c = \frac{N_t}{N_d} \quad (1)$$

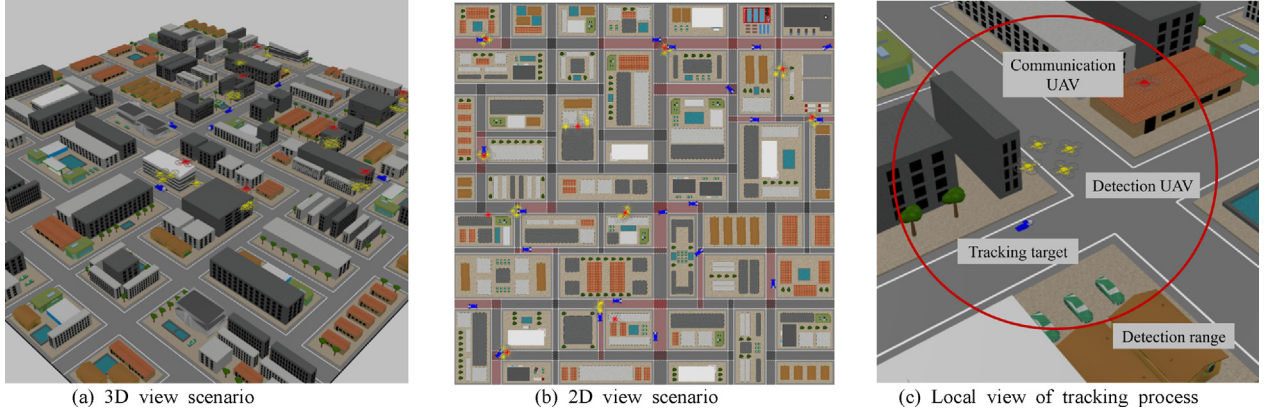


Fig. 1 Collaborative detection and tracking simulation scenario of UAVS.

Table 1 SoS indicators for UAVS typical scenario.

Type	Category	SoS indicator
UAVS	Detection UAV	Quantity, Maximum speed, Maximum acceleration, Detection (Communication) range
	Communication UAV	
Target	Ground vehicle	

Besides the task completion rate, the number of disabled drones caused by targets' attack should also be considered in the effectiveness calculation. The disabled degree M_{dis} can be expressed as the ratio of the number of disabled drones to all drones. Combining two factors, the SoS effectiveness E_s of UAVS can be calculated by applying a weighting method as

$$E_s = \omega_1 M_c + \omega_2 (1 - M_{dis}) \quad (2)$$

where ω_1 and ω_2 are weights of the task completion rate and disabled rate. The value of weights is determined based on operational requirements from experts, where task completion rate is prioritized to reflect mission success as the primary objective, and disabled degree is considered to capture the system's survivability. Therefore, ω_1 and ω_2 are set to 0.8 and 0.2.

Utilizing the established SoS indicators and effectiveness measurement method, the SoS dataset can be acquired by the UAVS simulation system to build the Data-driven evaluation network, which will be discussed in the next section.

3.3. Data-driven SEE framework for UAVS

The proposed Data-driven SEE framework, shown in Fig. 2, consists of three main parts: UAVS simulation and data processing, the Data-driven effectiveness evaluation network, and active learning for sample selection.

- (1) UAVS simulation and data processing. Data is the core of the Data-driven SEE method. All the data for training the evaluation model are derived from the dataset collected by the UAVS simulation system. In this module, simulation data of UAVS with important SoS indi-

cators are sampled from the data sampling space and then simulated through the simulation system. The effectiveness calculated by simulation would be collected and combined with the indicators to become the dataset for the evaluation module.

- (2) Data-driven effectiveness evaluation network. Based on the simulated dataset, this part establishes and trains a deep learning neural network for SEE of UAVS. In response to the difficulty of deep learning methods in directly matching the indicator structure of swarm systems with practical application requirements, an evaluation network based on a residual neural network and a skewed loss function is designed. After fully training the Data-driven model, the UAVS system with certain indicators can be quickly evaluated through this model.
- (3) Active learning for sample selection. The Data-driven evaluation model heavily relies on the simulation data. However, the collection of datasets requires the simulation-based labeling of unlabeled indicators' data, which would cost a significant amount of time to execute. To reduce the high-cost simulation annotation, an active learning strategy is constructed to screen a small number of high-value unlabeled samples from the sampling space. By simulating these limited but high-value samples, the number of simulations can be reduced while ensuring accuracy.

Moreover, the proposed framework is versatile and can be applied to UAVS with different indicators under various scenarios, not limited to the mentioned tracking mission. The framework is adaptable, enabling its use for different types of UAVS applications, such as surveillance, search and rescue, and environmental monitoring, by adjusting the relevant SoS indicators to meet the specific needs of each scenario.

4. Data-driven evaluation network for UAVS

4.1. Framework of proposed SEE method

The framework of the proposed Data-driven effectiveness evaluation network, named ResNet with Skewed Loss (ResNet-Skew), is depicted in Fig. 2. The network model con-

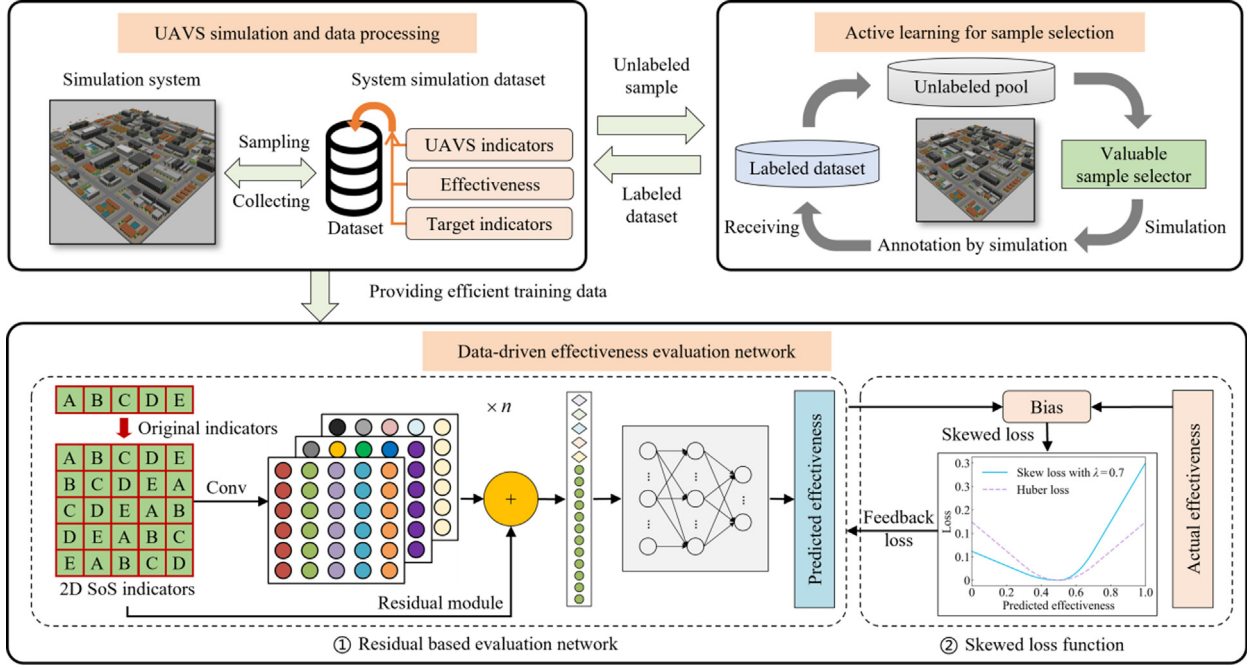


Fig. 2 Data-driven SEE framework for UAVS.

sists of three components. First, a feature transformation method based on loop stacking is introduced to convert the SoS indicators of UAVS into a two-dimensional format. Next, convolutional and residual structures, along with fully connected layers, are employed to extract relevant information and calculate the effectiveness value. Additionally, to better align Data-driven models with real-world applications, this study designs a skewed loss function to relieve overestimated predictions during training. Detailed descriptions of each module will follow.

4.2. Residual-based evaluation network

Deep learning methods are highly effective at simulating and fitting complex systems, but they are typically applied to multi-dimensional data. In SEE, the SoS indicators are one-dimensional, necessitating a mechanism to transform indicators into a suitable format. Transforming one-dimensional data into two-dimensional structures has been widely used to leverage the strong representation capabilities of convolutional neural networks in domains where data lacks inherent spatial organization. Prior works in transformed one-dimensional learning domains have shown that reshaping or projecting 1D vectors into 2D indicators allows CNNs to better capture local patterns and feature interactions.^{41,42} When the number of features is large, simple reshaping can create sufficiently rich 2D layouts for convolution. However, in scenarios like our SEE task, where the SoS indicator vector contains 12 features, such reshaping is limited in its ability to provide enough space. Therefore, a stacking-based transformation can be employed to construct a two-dimensional structure suitable for convolution.

To address this, we propose a loop stacking approach that performs cyclic shifts of the original feature vector, generating a series of rotated variants. These indicators are stacked row-

wise to form a structured 2D matrix. Compared to direct duplication or stacking, this method increases the effective receptive field, enabling the CNN to learn richer feature relationships without increasing input dimensionality or requiring explicit spatial encoding. In this SEE problem, the original 12 SoS features are transformed to 12×12 feature map and then are processed by convolutional architectures, allowing for the exploration of feature correlations and uncovering deep interconnections between features.

To clearly illustrate the advantages of the loop stacking method in processing 2D system indicator features, a comparison is conducted between this method and the direct stacking method in Fig. 3, highlighting the differences in processing one-dimensional data with five features. Fig. 3(a) shows the illustration that a 3×3 convolution structure is applied to the indicators processed by different stacking methods. Fig. 3(b) shows the times of each indicator that this convolution can include. Features processed by the loop stacking method can simultaneously operate on five features in one kernel. Conversely, the direct stacking convolution structure can only extract and process information from three adjacent elements. The loop stacking can effectively extract the interrelated information, enhancing the field of view without altering the convolution kernel size, thus yielding more valuable insights and more comprehensively unveiling the interrelationships among factors.

To extract potential information and interrelations among the two-dimensional features, a SEE model based on the ResNet network⁴³ is utilized as the backbone to predict the effectiveness of UAVS. ResNet model's structure is selected and modified for its ability to handle deeper architectures while preserving important information through residual connections, which mitigates the vanishing gradient problem. Its convolutional layers are capable of extracting both local and global patterns from the transformed bi-dimensional data,

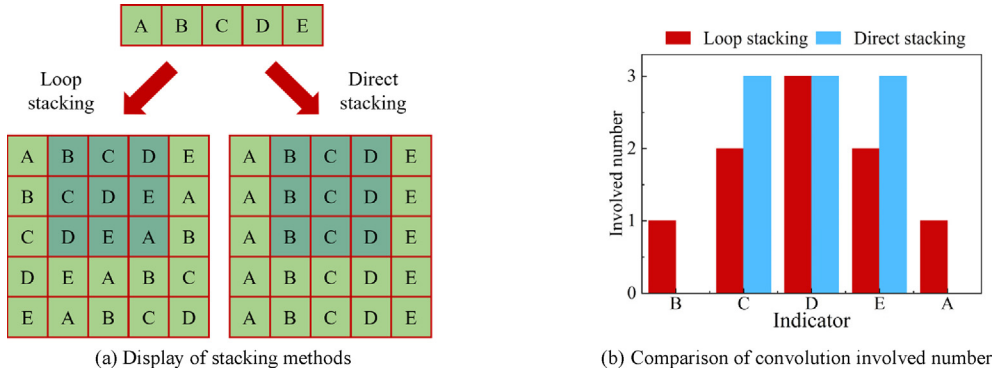


Fig. 3 Comparison between loop stacking and direct stacking.

Table 2 Structure of ResNet-Skew.

Block	Structure	Block	Structure
Block 1	$[3 \times 3, 64] \times 1$	Block 2	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$
Block 3	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$	Block 4	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$
Block 5	Average pooling	Block 6	$[64, 64]$

which is crucial for capturing the complex relationships among SoS indicators. The detailed structure of the proposed ResNet-Skew model is shown in Table 2.

The model consists of 13 convolutional layers, each with 3×3 kernels, that process the SoS indicator features. Residual structures are incorporated into the network, allowing shallow features to skip two convolution layers and reach deeper layers in blocks 2, 3, and 4, effectively maintaining robust training performance even as the network depth increases.

In terms of the residual process, denote the original data as X , and the data processed by the first and the second convolution layer of blocks 2 to 4 as Z and Y , which can be expressed as

$$\begin{cases} Z = \text{ReLU}(\text{BN}(W_1 \otimes X)) \\ Y = \text{ReLU}(\text{BN}(W_2 \otimes Z)) + W_3 \otimes X \end{cases} \quad (3)$$

where ReLU is the rectified linear unit and BN denotes batch normalization and W_1, W_2 denote the kernel of convolution layer. W_3 is a linear transformation matrix. $\otimes$ represent the convolution process. After two convolution layers, the original feature X would be directly added to the processed data as Y .

The goal of the SEE model is to predict the effectiveness value. Therefore, the feature data processed by the convolution and residual structures needs to be transformed into a one-dimensional effectiveness value. After data conversion by average pooling, two Fully Connected (FC) layers are used to regress the SoS effectiveness. The output data x_o is transformed into a 0–1 effectiveness value E_s using the Sigmoid function as

$$E_s = \frac{1}{1 + e^{-x_o}} \quad (4)$$

4.3. Skewed loss function

The loss function is essential for assessing the accuracy of a model's predictions, quantifying the discrepancy between the model's predicted value $f(x_i)$ and the true effectiveness value y_i . A smaller loss value indicates better model robustness. For this task, the Huber loss function⁴⁴ is used as the foundation due to its combination of the advantages of both the Mean Absolute Error (MAE) and Mean Squared Error (MSE) loss functions. The Huber loss applies the MAE for large errors, using a constant gradient for training, while it switches to the MSE for small errors, allowing for a smaller gradient and ensuring that the loss decreases as the error diminishes. Let x_i be a sample with a true label y_i and predicted value $f(x_i)$. The corresponding Huber loss is defined as

$$L_{\text{Huber}} = \begin{cases} 2\delta|f(x_i) - y_i| - \delta^2, & |f(x_i) - y_i| > \delta \\ (f(x_i) - y_i)^2, & |f(x_i) - y_i| \leq \delta \end{cases} \quad (5)$$

where δ is the threshold parameter. When the bias between $f(x_i)$ and y_i smaller than or equal to δ , the Huber loss applies MSE (squared error), and switches to MAE (absolute error) when the error exceeds δ . This design allows the loss to be quadratic near zero to ensure smooth gradients, and linear for large errors.

However, in practical scenarios, particularly when evaluating the effectiveness of swarm systems, discrepancies where the model's predicted effectiveness exceeds the actual effectiveness can result in an overestimation of the system's capabilities. If this overestimation (predicted effectiveness exceeding actual performance) leads to deployment, the system may fail during execution, as its actual performance will not meet expectations. Such misalignments can have severe consequences, particularly

in critical missions, where failure due to underperformance may lead to catastrophic outcomes. On the contrary, when the predicted effectiveness is lower than the actual effectiveness, the system may be overly cautious, but the consequences are generally less severe. Underestimation typically leads to inefficiencies or suboptimal performance, without the high-risk potential associated with overestimating system capabilities.

In real-world applications, particularly in UAVS tasks, the cost of failure resulting from overestimation is far more damaging than the inefficiencies caused by underestimation. Unfortunately, standard prediction models do not account for these critical differences in how various types of prediction errors impact system performance. To address this issue, we modify the Huber loss function by introducing a skewed loss function. This adjusted function places greater emphasis on cases where the predicted effectiveness exceeds the actual effectiveness, thus reducing the likelihood of deploying systems with overly optimistic predictions that could result in task failure. The skewed loss function is defined as

$$L_{SL} = \begin{cases} e^{\lambda}(2\delta|f(\mathbf{x}_i) - y_i| - \delta^2), & \delta < f(\mathbf{x}_i) - y_i \\ e^{\lambda}(f(\mathbf{x}_i) - y_i)^2, & 0 < f(\mathbf{x}_i) - y_i \leq \delta \\ e^{-\lambda}(f(\mathbf{x}_i) - y_i)^2, & -\delta < f(\mathbf{x}_i) - y_i \leq 0 \\ e^{-\lambda}(2\delta|f(\mathbf{x}_i) - y_i| - \delta^2), & f(\mathbf{x}_i) - y_i \leq -\delta \end{cases} \quad (6)$$

where λ represents a penalty adjustment parameter, with a value range of $[0, 1]$.

This modified loss function dynamically adjusts its magnitude based on the parameters δ and e^{λ} . The loss function is divided into four segments based on the prediction error. When the model's predicted value exceeds the actual label, a higher penalty weight e^{λ} is applied, as illustrated in the first two cases of Eq. (6). In contrast, when the predicted value is lower than the actual value, a smaller penalty weight $e^{-\lambda}$ is used, with the switching behavior of the MSE and MAE loss functions still governed by δ . The latter two cases of Eq. (6) correspond to this scenario. Notably, when $\lambda = 0$, the exponential weights reduce to 1 and the loss becomes equivalent to the standard Huber loss, which treats overestimation and underestimation symmetrically.

The proposed skewed loss is a risk-aware generalization of Huber loss, where λ controls the degree of asymmetry in penalizing overestimation. This design is motivated by the asymmetric risk in UAVS mission, where overestimation of effectiveness can result in mission failure, while underestimation typically leads to conservative but safer decisions. Traditional symmetric losses like MSE or Huber do not capture this risk profile. The skewed loss is theoretically supported by cost-sensitive learning and quantile regression,⁴⁵ which introduce asymmetry into learning objectives to reflect real-world decision priorities.

Fig. 4 illustrates the impact of varying the λ parameter on the skewed loss function at an actual effectiveness value of 0.5. The dashed line represents the loss function values using the original Huber loss, while the other curves show the skewed loss function with different values of λ . On all curves, the gradient decreases as the error diminishes and stabilizes at larger errors, allowing for variable gradient strategies based on the error magnitude. As λ increases, the penalty for samples with predicted values above the actual values becomes more

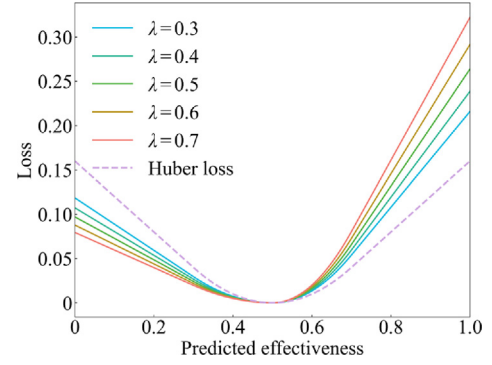


Fig. 4 Loss curve of skewed Huber loss with effectiveness label 0.5.

pronounced, as observed in the right-hand side of the loss function. This adjustment mitigates the adverse effects of overestimated predictions on decision-making.

The skewed loss maintains the robustness of the Huber formulation (i.e., quadratic loss for small errors and linear loss for large ones), while adding directional sensitivity through λ . This provides a flexible, interpretable, and tunable loss function that aligns optimization objectives with real-world safety and robustness needs in UAVS systems. As shown in Section 6.5, increasing λ leads to a consistent reduction in overestimation bias, while maintaining comparable RMSE. This demonstrates the practical effectiveness of the skewed loss in balancing performance accuracy with risk sensitivity. Section 6.5 will discuss the effect of skewed loss in detail.

The residual-based evaluation network and the skewed loss function module together form the ResNet-Skew SEE framework of UAVS. This model enables rapid and accurate predictions of SoS effectiveness and adapts to the real-world demands of UAVS tasks.

5. Active learning for sample selection

Training a Data-driven SEE model requires a large amount of labeled data. However, obtaining effectiveness values for UAVS with different indicators via simulations is both time-consuming and costly. Therefore, it is crucial to minimize training costs by selecting high-value samples. Active learning offers a solution by selecting the most informative samples from an unlabeled dataset for annotation, thus reducing the overall number of required annotations during training.⁴⁶ By incorporating active learning into the SEE framework, we can effectively reduce sample annotation cost.

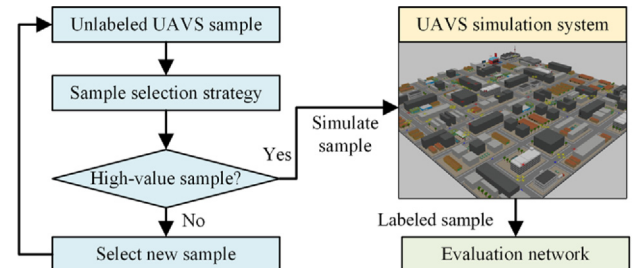


Fig. 5 Application process of active learning method in SEE.

5.1. Framework of active learning

The active learning framework is shown in Fig. 5. Initially, an unlabeled set of SoS indicator samples is generated. The active learning method evaluates each sample's value and selects the most informative ones. These high-value samples are then annotated by calculating their SoS effectiveness using the simulation system. As a result, only a small portion of the total sample space is used, reducing annotation costs while ensuring good model generalization.

The problem of selecting initial sample annotations is a sample selection issue within unsupervised learning. Sample selection in active learning typically focuses on two criteria: representativeness and diversity.⁴⁷ Representativeness ensures that the selected samples reflect the entire sample space, while diversity ensures that samples are evenly spread across the space and not clustered in a limited region. These criteria are critical in selecting initial samples for training the neural network.

Given the complexity of both the UAVS system and the neural network framework, the initial sample size required for this problem is larger than traditional active learning regression tasks, ranging from hundreds to thousands of instances. Therefore, we propose an unsupervised active learning strategy, iRDM-RGS, designed to handle large-scale sample selection. This strategy utilizes both representativeness and diversity criteria and is based on iterative Representativeness-Diversity Maximization (iRDM)⁴⁷ and random greedy sampling, as detailed in Algorithm 1.

For the initial unlabeled dataset $\mathbf{O}$, the proposed iRDM-RGS algorithm first selects a few representative and diversity-based samples using the iRDM method (lines 1–9). Subsequently, the algorithm employs a random greedy strategy grounded in diversity criteria (Algorithm 2) to select the remaining samples, culminating in the final sample set $\mathbf{X}$. After the sample selection is complete, the algorithm calculates the effectiveness label values Y for the sample set using the UAVS simulation system, ultimately yielding a labeled dataset of SoS indicators $\mathbf{D} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_M, y_M)\}$ for deep learning model training. The following sections will detail the two main modules of this algorithm: the iRDM based initial sample selection and the random greedy sampling strategy.

5.2. iRDM based initial sample selection

For the initial unlabeled dataset $\mathbf{O}$, the algorithm first applies K-means clustering to initially select samples that are highly representative and diverse. Clustering, a frequently used method in active learning, can identify samples that are both representative and diverse.¹² The number of clusters is critical for the sample selection process in clustering algorithms. Choosing a large number of clusters may lead to local optima instead of global optima, complicating the algorithm's search for the true clustering structure. Given the large sample size in the SEE problem, the algorithm sets the number of clusters to $N = 200$, assuming the maximum number of labeled samples M is between 200 and 1000. Within each cluster, the sample closest to the center is chosen as the candidate to represent the cluster.

Algorithm 1. Main framework of iRDM-RGS.

Input: Unlabeled sample sets: $\mathbf{O} = \{\mathbf{O}_1, \mathbf{O}_2, \dots, \mathbf{O}_K\}$, Number of the unlabeled samples: K , Budget of sample selection: M , Iteration number: $G_{\max}$, Number of sample selected by iRDM: N .
Output: Training dataset with label.

1. Perform K-means cluster to $\mathbf{O}$ (N classes), each cluster center would be denoted as $\mathbf{C}_p = \{\mathbf{C}_{p1}, \mathbf{C}_{p2}, \dots, \mathbf{C}_{pN}\}$.
2. Select the sample closest to each center as the candidate set $\mathbf{O}_c = \{\mathbf{O}_{c1}, \mathbf{O}_{c2}, \dots, \mathbf{O}_{cN}\}$.
3. $g \leftarrow 0$.
4. **for** $g < G_{\max}$ **do**
5. **for** $\mathbf{O}_{cl}$ in $\mathbf{O}_c$ **do**
6. Replace the sample $\mathbf{O}_{cl}$ with the sample behaving the highest value according to Eq. (9).
7. **end for**
8. $g \leftarrow g + 1$.
9. **end for**
10. Taking samples in $\mathbf{O}_c$ as the foundation, select the remain $M - N$ samples $\mathbf{X}_r = \{\mathbf{x}_{N+1}, \mathbf{x}_{N+2}, \dots, \mathbf{x}_M\}$ through the random greedy sampling strategy (Algorithm 2).
11. Receive the finally selected sample sets $\mathbf{X} = \mathbf{O}_c \cup \mathbf{X}_r = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_M\}$.
12. Simulate the sample in $\mathbf{X}$ to receive the effectiveness Y .
13. **return** $\mathbf{D} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_M, y_M)\}$.

To improve both representativeness and diversity of the selected samples from the candidate set $\mathbf{O}_c$, this study applies the iRDM method to update samples within each cluster, as described in lines 4 through 9 of Algorithm 1. The selection of representative samples within cluster $\mathbf{C}_l$ is guided by both diversity and representativeness criteria. The representativeness of a sample is measured by its average distance to other samples in the same cluster. For a cluster $\mathbf{C}_l = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{P_l}\}$, P_l is the number of samples in this cluster. The representativeness score of the sample $\mathbf{x}_i$ is calculated as

$$R(\mathbf{x}_i) = \frac{1}{P_l - 1} \sum_{\mathbf{x}_k \in \mathbf{C}_l} \|\mathbf{x}_i - \mathbf{x}_k\| \quad (7)$$

The diversity criterion evaluates the minimum distance from each sample $\mathbf{x}_i$ in cluster $\mathbf{C}_l$ to the representative samples $\mathbf{O}_{ck}$ in other clusters, as

$$D(\mathbf{x}_i) = \min_{\substack{k \in [1, N] \\ k \neq l}} \|\mathbf{x}_i - \mathbf{O}_{ck}\| \quad (8)$$

To simultaneously enhance both diversity and representativeness, we maximize inter-cluster diversity score while minimizing the intra-cluster representativeness score. For each sample in cluster $\mathbf{C}_l$, we calculate $R(\mathbf{x}_i)$ and $D(\mathbf{x}_i)$, and select the optimal sample $\mathbf{x}_i^*$ in cluster $\mathbf{C}_j$ using the following formula:

$$\mathbf{x}_i^* = \arg \max_{i \in [1, P_l]} [D(\mathbf{x}_i) - R(\mathbf{x}_i)] \quad (9)$$

The selected sample $\mathbf{O}_{cl}$ in the cluster is then replaced with $\mathbf{x}_i^*$, and the update is performed for all samples within the cluster.

This selection process is illustrated in Fig. 6(a). A , B , and C represent candidate points within each cluster. During the iteration, each point in the cluster is assessed according to the formula in Eq. (9). For example, point D 's representativeness score $R(\mathbf{x}_D)$ is calculated by measuring its average distance to other points in the cluster, shown by three short dashed

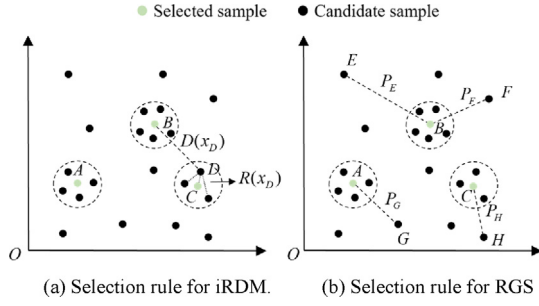


Fig. 6 Selection rules of iRDM-RGS.

lines. Additionally, the distance from D to B measures the diversity $D(x_D)$, indicating the minimal distance to other cluster centers. The combined evaluation of these two metrics determines whether point D should replace the old selected sample point C .

To ensure the stability of the sample updates, the aforementioned method is iterated $G_{\max}$ times. The updated sample set $O_c = \{O_{c1}, O_{c2}, \dots, O_{cN}\}$ is obtained. Based on this dataset, a random greedy strategy based on diversity criteria would be further employed to select the remaining samples.

5.3. Random greedy sampling strategy

After applying the iRDM method, the sample set O_c consists of N selected samples, leaving $M - N$ samples unselected. To further enhance the diversity of the selected samples and ensure a broad representation of the sample space, we introduce the Random Greedy Sampling (RGS) strategy. This strategy combines greedy selection with randomness, as described in Algorithm 2. iRDM primarily focuses on selecting a small number of highly representative samples, and the RGS strategy aims to enhance the diversity of the sample set by selecting additional samples that are distant from those already chosen, thereby ensuring broader coverage of the sample space. This shift in focus allows RGS to complement iRDM by filling the gaps left by the representativeness-driven initial selection, thus optimizing the overall sampling process.

Algorithm 2. Random greedy sampling strategy.

Input: Unlabeled sample sets: $O = \{O_1, O_2, \dots, O_K\}$, Sample selected by iRDM: $O_c = \{O_{c1}, O_{c2}, \dots, O_{cN}\}$, Budget of sample selection in RGS: $M - N$.
Output: Samples selected by RGS: $X_r = \{x_{N+1}, x_{N+2}, \dots, x_M\}$.
1. Count all the remaining unlabeled sample set:
 $R = O - O_c = \{R_1, R_2, \dots, R_{K-N}\}$.
2. $j \leftarrow 0$.
3. **while** $j < M - N$ **do**
4. **for** R_i in R **do**
5. Calculate the minimum sample distance from R_i to O_c by Eq. (10).
6. **end for**
7. Calculate the selection probability of samples by Eq. (12).
8. Use the roulette wheel method to select a sample R_i from R to enter the set O_c based on the probability.
9. Remove R_i from set R .
10. $j \leftarrow j + 1$.
11. **end while**
12. Set $X_r \leftarrow O_c$.
13. **return** Sample selected by RGS: $X_r = \{x_{N+1}, x_{N+2}, \dots, x_M\}$.

After clustering iteration yields the sample set O_c , this algorithm tallies the remaining samples into R and chooses $M - N$ samples from this set. This approach is conceptually similar to a subset selection problem, which can be tackled using models and solution strategies from the latter. According to the principle of subset selection, we aim to include samples that are more distant from the existing set, thus broadening the sample space. However, solving this problem with exact optimization methods is computationally expensive, particularly for large-scale datasets. To address this, the algorithm employs a greedy approach to iteratively select samples.

For each sample point x_i in R , the distance to the currently selected set O_c is computed, with the minimum distance being used as the selection criterion. The minimum distance is calculated as

$$C(x_i) = \min_{x_j \in O_c} \Delta(x_i, x_j) \quad (10)$$

A greater minimum distance indicates that the sample is more distant from the distribution space of O_c , thus enhancing diversity. Normally, a greedy optimization would select the sample with the maximum $C(x_i)$ to add to the set O_c . This selection criterion can be represented as

$$u = \arg \max_{x_i \in R} C(x_i) \quad (11)$$

However, greedy methods are highly sensitive to the initial sample selection, and inappropriate initial samples can lead the algorithm to local optima. To mitigate this, a randomized greedy approach is introduced. Rather than deterministically choosing the sample with the maximum distance, a probability is assigned to each sample based on its distance. This probabilistic approach reduces the likelihood of the algorithm becoming stuck in local optima. The probability is calculated as

$$p_i = C(x_i)^\gamma / \sum_{j=1}^{M-N} C(x_j)^\gamma \quad (12)$$

where γ is a greediness control parameter. When $\gamma = 0$, all samples have an equal probability of being selected, while increasing γ skews the probability towards samples with larger distances. As γ approaches infinity, the model closely approximates pure greedy selection.

Using this probabilistic model, the roulette wheel method is applied to select samples based on their assigned probabilities, which are then added to O_c . This process continues until the sample set reaches the budget size M .

The selection rule of RGS is illustrated in Fig. 6(b). Suppose A , B , and C represent samples selected by iRDM. During the greedy selection process, each candidate point in the sample space is evaluated based on its minimal distance to the selected points, which is converted into a probability. For example, points E , F , G , and H are evaluated, and their probabilities P_E , P_F , P_G , and P_H are computed accordingly. The next sample is selected based on these probabilities.

Selecting samples using the iRDM-RGS strategy, the algorithm then computes the effectiveness values and ultimately generates a labeled dataset for deep learning model training. The combination of iRDM's focus on representativeness and RGS's emphasis on diversity ensures efficient sample selection,

reduces annotation costs and enhances model training efficiency for large-scale active learning tasks.

6. Experiments and analysis

This experiments section will test the Data-driven evaluation network and the corresponding active learning. All algorithms in this section are programmed using Python 3.8 on the Win10 64 bit operating system with Intel Core i9 13 900 k CPU@3.00 GHz, 64 GB RAM and NVIDIA GeForce RTX 4070Ti GPU hardware environment.

6.1. UAVS mission scenario and data processing

Given the difficulty and high cost of collecting real operational data for large-scale UAVS systems, the simulation system can provide a credible and practical alternative, ensuring that the data reflects the uncertainties and dynamics of real-world scenarios. Collect data samples using the constructed simulation system for UAVS collaborative detection and tracking tasks. Based on simulation scenario information and preliminary experiments, a total of 12 SoS indicators of UAVS and targets will be sampled and incorporated into a dataset within a defined range. To ensure the accuracy of the effectiveness evaluation model, it is essential to obtain a uniformly distributed set of SoS indicator combinations that cover all possible scenarios in the evaluation process. Given the wide range of potential values for the 12 SoS indicators, a larger sampling space is necessary to ensure comprehensive coverage. To address this, the Latin hypercube sampling method is employed to collect 10 000 initial sample points. This method guarantees that the sample points are well-distributed across the entire parameter space, reducing the risk of clustering around certain values and maximizing scenario representation. These unlabeled samples will be further used for simulating, training, and validating both the Data-driven SEE model and active learning strategy.

Each UAVS configuration is simulated when the corresponding data is selected as the training dataset. The simulation environment incorporates multiple stochastic factors, including communication interference, detection uncertainty, and probabilistic adversarial strikes. These factors introduce dynamic perturbations and randomness into the simulation process, making the outcomes more representative of real-world UAVS operations. Each configuration is tested through ten repeated simulation runs, thereby capturing the expected performance while smoothing out occasional fluctuations due to environmental noise. This approach allows the dataset to reflect both the variability and the stability of system behavior under uncertain operational conditions.

6.2. Data-driven evaluation network performance

6.2.1. Experimental settings and metrics

To evaluate the performance of the proposed Data-driven SEE model, ResNet-Skew, 2000 samples are randomly selected from 10 000 unlabeled data points and annotated using the simulation system. These samples are split into a training set of 1000 samples and a testing set of 1000 samples. The decision to use 1000 samples for training is aimed at minimizing the computational cost of obtaining labeled data through simulations. By keeping the training set smaller, the focus remains on

efficient model development, while the testing set, also containing 1000 samples, is used to ensure a robust evaluation across a wider range of scenarios. This helps to assess the model's performance and generalization capability in diverse situations.

The hyperparameters of the model are shown as follows: The model is trained for 50 epochs, with optimization carried out via Stochastic Gradient Descent (SGD), and an initial learning rate of 0.01, which is reduced by 10 % after the 10th and 30th epochs. For the skewed loss function, the parameters λ and δ are set to 0.7 and 0.05, respectively. The performance is evaluated using the Root Mean Square Error (RMSE) and MAE metrics to gauge prediction accuracy.

For comparison, the proposed ResNet-Skew algorithm is compared with six Data-driven evaluation methods proposed in recent years and state-of-the-art deep learning algorithms:

- (1) SE-DEA-SVM.²⁹ This method applies a Support Vector Machine (SVM) as an algorithm to evaluate and predict the effectiveness of configuration schemes processed by data-elevation analysis. The SVM will be directly applied to 1D SoS indicators.
- (2) KL-SAE.¹⁰ This method uses a sparse autoencoder based on KL divergence for the problem of damage effectiveness evaluation. In KL-SAE, the hidden layer features of the encoder, which contribute most to data reconstruction, are selected, and features that contribute least are abandoned. KL-SAE will be directly applied to 1D SoS indicators.
- (3) 1D-CNN.³⁵ This method employs a parallel one-dimensional convolutional neural network for addressing the problem of feature extraction in SEE. In 1D-CNN, the 1D system indicators are processed by 1D convolutional structures with three different kernel sizes. The extracted features are synthesized to extract potential information and predict the effectiveness by DNN. 1D-CNN will also be directly applied to 1D SoS indicators.
- (4) BPNN-ATT.³⁰ This method employs a backpropagation neural network as a baseline model to calculate the effectiveness of UAVS. In BPNN-ATT, a feature extraction method is proposed based on a feedback closed-loop strategy, which extracts features from the original data space in reverse, based on dimension reduction by principal component analysis. The feature extraction would be applied to select SoS indicators, and BPNN will be trained to predict the effectiveness.
- (5) Graph Convolution Networks (GCN).³⁷ GCN is a state-of-the-art deep learning framework designed to process graph-structured data by iteratively aggregating and transforming information from neighboring nodes to capture complex relational dependencies. In this experiment, the GNN model treats each SoS indicator as a node in a graph with fully connected edges.
- (6) Transformer.⁴⁸ Transformer is a state-of-the-art deep learning method. This model utilizes an encoder structure composed of multiple layers of multi-head self-attention and feed-forward networks. In this experiment, one-dimensional SoS indicators will be processed as a sequence by the Transformer.

The SE-DEA-SVM, KL-SAE, 1D-CNN, BPNN-ATT algorithms aim to address the SEE problem from three different perspectives: traditional machine learning, autoencoder,

and deep learning. GCN and Transformer are state-of-the-art deep learning algorithms in these years.

6.2.2. Model evaluation and comparison analysis

The proposed ResNet-Skew model is trained sufficiently to ensure convergence. Fig. 7 presents the RMSE convergence curves on both the training and test datasets during iterative optimization. The model's error decreases rapidly in the early stages of training. As the number of iterations increases, the RMSE curves of both the training and test sets gradually stabilize, indicating steady convergence. After sufficient training, the RMSE on the training set drops below 0.02, while the RMSE on the test set stabilizes around 0.08. The two curves exhibit similar convergence trends with no observable signs of overfitting, suggesting that the model achieves a good balance between fitting capacity and generalization.

Based on the training and testing sets described earlier, we train the ResNet-Skew model alongside several comparison algorithms. The experiment is repeated ten times to statistically assess the performance of each model using the RMSE and MAE metrics, with the results presented in Table 3. To provide a more robust statistical analysis (mean and standard deviation), the symbols “+” indicate that the ResNet-Skew model is significantly “superior/equivalent/inferior” to other algorithms based on the Wilcoxon rank-sum test at the 5 % significance level.

The results demonstrate that the ResNet-Skew model significantly outperforms the other algorithms across both RMSE and MAE metrics. The 1D-CNN algorithm performs the worst, with an RMSE exceeding 0.2, indicating substantial errors. In contrast, the proposed two-dimensional convolutional structure and following residuals structure of the ResNet-Skew model, which is designed for the 2D data, captures the interrelationships among the various indicators more effectively than the one-dimensional convolution used by the 1D-CNN. This deeper, multidimensional feature extraction enhances the model's performance. Furthermore, GCN and Transformer, which are two advanced deep learning models, also achieve limited performance. The relatively poor performance of GCN is primarily due to the absence of inherent graph structure in the SoS indicator data. The 12-dimensional system indicator vector represents parallel features without explicit topological relationships.

Similarly, Transformer is designed for long-sequence modeling tasks, where capturing long-range dependencies is essential. In our problem, the feature structure and limited dataset

size are insufficient to utilize the self-attention mechanism effectively. The traditional methods, such as SE-DEA-SVM and BPNN-ATT exhibit relatively higher prediction errors. SE-DEA-SVM lacks feature extraction, which limits its ability to capture complex nonlinear relationships among SoS indicators. BPNN-ATT performs better than SE-DEA-SVM but still lacks the depth and capacity to extract deep feature representations, leading to suboptimal accuracy. KL-SAE achieves improved performance by using an autoencoder structure for feature compression, but its unsupervised nature may not align well with the effectiveness prediction task, resulting in moderate errors compared to ResNet-Skew. The proposed deep learning structure of ResNet-Skew allows for better extraction of deep information from the SoS indicators of UAVS, enabling more accurate predictions through the integration of both potential features and the original features from the residual modules.

6.2.3. Model structure validation and analysis

Following model validation, this section assesses the effect of two main modules within the ResNet-Skew algorithm: the two-dimensional loop stacking of system indicators and the skewed loss function.

During the two-dimensional transformation of system indicators, this algorithm utilizes a method based on loop stacking. To validate the effect of 2D transformation and the loop stacking-based transformation methods, a comparison of different data processing methods is conducted. The 1D convolution with the same residual structure is employed to compare the effect of the 2D transformation. Furthermore, the direct stacking-based 2D transformation method described in Section 4.2 and Fig. 3(a) is also included to verify the advantage of loop stacking in 2D transformation. The results of this comparison are presented in Table 4.

While three models employ the same residual structure, the performance of applying loop stacking and direct stacking-based 2D transformation is superior to direct 1D convolution, which can demonstrate that the 2D transformation can extract more complex and higher-level information from the original SoS indicators. Furthermore, the evaluation network employing loop stacking also outperforms the algorithm with direct stacking modules in terms of both RMSE and MAE metrics. By intertwining variables across the entire two-dimensional metric structure, the convolutional module can effectively extract the correlations between different metrics, thus enhancing model performance.

Furthermore, to illustrate the impact of the skewed loss function on relieving overestimation, this section compares models using biased loss functions with varying parameters λ with those using Huber loss functions. Initially, to visually demonstrate the effect of the skewed loss function, we compare the overestimated bias (predicted values minus actual values) of some samples trained with the skewed loss function and the normal Huber loss function, as depicted in Fig. 8.

The red line between the Huber loss point (red point) and the skewed loss point (green point) represents the situation where the overestimated bias is alleviated by the skewed loss. The skewed loss function significantly reduces the overestimated bias for most samples compared to the red point of the Huber function. However, in some cases, there is a significant negative bias, where the original small loss is magnified

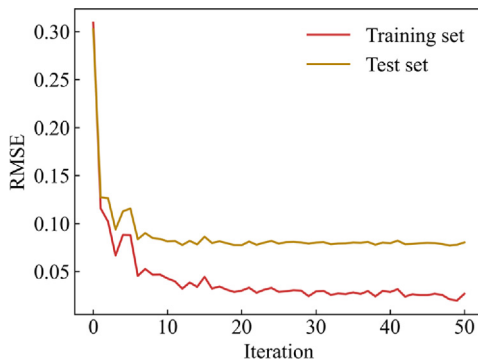


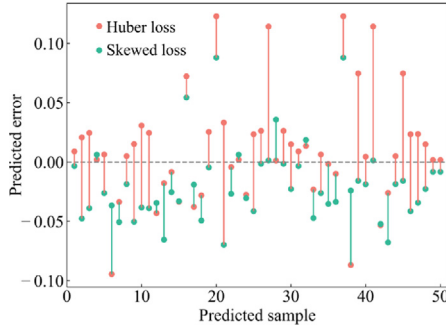
Fig. 7 RMSE iteration curves of ResNet-Skew.

Table 3 Statistical results for different SEE models.

Metrics		ResNet-Skew	SE-DEA-SVM	KL-SAE	1D-CNN	BPNN-ATT	GCN	Transformer
RMSE	Mean	0.081	0.181(+)	0.167(+)	0.232(+)	0.121(+)	0.230(+)	0.232(+)
	Std (10^{-3})	2.21	3.10	17.0	6.78	2.15	2.83	0.881
MAE	Mean	0.061	0.149(+)	0.134(+)	0.199(+)	0.090(+)	0.199(+)	0.198(+)
	Std (10^{-3})	2.00	2.20	14.5	6.65	7.92	3.40	2.15

Table 4 Statistical results for different processing methods.

Strategy	RMSE		MAE	
	Mean	Std (10^{-3})	Mean	Std (10^{-3})
Loop stacking	0.081	2.21	0.061	2.00
Direct stacking	0.090(+)	2.01	0.070(+)	2.17
1D with residual	0.108(+)	5.00	0.076(+)	2.33

**Fig. 8** Comparison of predicted error values under different loss functions.

after applying the skewed loss function. This shift towards negative bias is a natural consequence of introducing the skewed loss, as it aims to reduce overestimated bias at the cost of slightly increasing the underestimation for a few samples.

Since the skewed loss can cause a negative shift in model loss, it is crucial to balance the degree of skew to optimize both bias reduction and overall model performance. Too much skew may reduce overestimation but increase RMSE, so finding an appropriate parameter λ is crucial. Therefore, we conduct experiments comparing the performance of skewed loss functions with different parameters λ . The proportion of samples with predicted values greater than actual values and the corresponding RMSE indicators are statistically calculated as metrics. The results are shown in Table 5. The overestimated bias represents the proportion of samples with overestimated bias, while the large overestimated bias indicates the percentage of samples where the overestimated bias between the predicted performance value and the actual value exceeds 0.1.

In Table 5, as the parameter λ increases, the proportion of samples with overestimated and large biases decreases, but the RMSE index increases. Specifically, when $\lambda = 0$, the skewed loss reduces to the standard Huber loss, treating overestima-

Table 5 Bias and metrics under different skewed loss.

Parameter	Overestimated bias (%)	Large bias (%)	RMSE
$\lambda = 0.9$	22.60	8.60	0.086
$\lambda = 0.7$	25.20	8.14	0.081
$\lambda = 0.5$	32.70	15.30	0.08
$\lambda = 0.3$	36.60	16.70	0.077
$\lambda = 0$	42.60	19.30	0.076

tion and underestimation equally. This case serves as a natural baseline for comparison. In terms of RMSE, the model trained with $\lambda = 0$ achieves the best result (0.076), indicating that symmetric penalty functions may slightly improve general error minimization. However, it also results in the highest overestimation ratio (42.6 %) and the largest average overestimation bias. This suggests that while Huber loss provides good overall fitting accuracy, it lacks control over the directionality of prediction errors, especially those that are risky in real-world UAVS deployments.

Subsequently, when λ increases to 0.7, the sample bias is substantially reduced, with the number of overestimated bias samples falling below 10 %, and only a minimal increase in RMSE, compared to $\lambda = 0.5$. However, further increases in λ to 0.9 do not significantly alter the proportion of overestimated bias samples. Thus, $\lambda = 0.7$ strikes a good balance between reducing the overestimated bias and keeping RMSE within acceptable limits, ensuring that the model remains both accurate and efficient. This balance is significant because the goal is to minimize the impact of overestimation on decision-making, which is crucial in real-world applications. The slight increase in RMSE when using $\lambda = 0.7$, is a reasonable trade-off for the reduction in overestimated bias. This property is crucial in mission-critical UAVS tasks, where the cost of optimistic predictions can outweigh the cost of conservative ones. Thus, compared to the symmetric Huber loss, our skewed loss

provides a more suitable risk-sensitive objective for effectiveness estimation.

Based on the experiments, it is evident that the ResNet-Skew algorithm can effectively evaluate the SoS effectiveness of UAVS, outperforming other commonly used algorithms in terms of both RMSE and MAE indicators. The model's architecture enhances its metric performance and satisfies the actual application demand of the Data-driven SEE model.

6.3. Active learning strategy performance

This section employs the proposed active learning method iRDM-RGS to select high-value initial samples, validating the enhancement of active learning strategies on model training.

6.3.1. Experimental settings and metrics

To simulate an active learning scenario, we first annotate 1000 samples from the previously collected 10 000 unlabeled samples as the test set. The remaining 9000 samples are then subjected to active learning methods, selecting between 200 and 1000 samples for simulation-based annotation, which will serve as the training set for the model.

For the active learning model, the following experimental parameters are set after extensive testing: the iteration count $G_{\max}$ is set to 10, the RGS parameter γ is set to 20, and the sample collection size N through iRDM is set to 200. The RMSE and MAE metrics are used to assess the discrepancies between the model's predicted and actual values. Euclidean distance is chosen to calculate the sample values as it effectively measures spatial relationships between samples, especially for evenly distributed data. After standardization, it accurately reflects similarities between samples without being influenced by feature scale differences.

To comparatively validate the iRDM-RGS method, three recent active learning regression algorithms and random sampling method are selected for comparative analysis: (A) GSx,⁴⁹ Greedy sampling on the Inputs; (B) iRDM,⁴⁷ iterative representativeness-diversity maximization; (C) RD,⁵⁰ a method synthesized representativeness and diversity; (D) RS, a baseline method by randomly sampling. By integrating the aforementioned four methods with the Data-driven framework, we can verify the effect of these methods.

6.3.2. Model evaluation and algorithm comparison

We first evaluate the performance of the active learning algorithm by selecting 1000 samples from the 9000 unlabeled sam-

ples. The results for various active learning strategies are shown in Table 6.

Table 6 presents both the mean and standard deviation of RMSE and MAE obtained from multiple independent experiments to assess the performance stability of each active learning strategy. In Table 6, the iRDM-RGS active learning strategy outperforms all other methods in both RMSE and MAE, as well as showing superior stability. Compared to Random Sampling (RS), every active learning algorithm demonstrates significant improvements in the quality of initial sample selection, highlighting the impact of active learning on model performance. The performance of iRDM and GSx is quite similar, though GSx exhibits greater stability. This could be attributed to iRDM's tendency to select samples with decreasing representativeness and stability as the number of clusters grows. In contrast, iRDM-RGS combines clustering-based selection with greedy sampling to balance representativeness and diversity, achieving superior performance, particularly during large-scale sample selection.

To comprehensively evaluate the active learning methods, we would compare their performance across different sample selection sizes by selecting between 200 and 1000 samples (every 200 samples) from 9000 unsupervised samples. This more extensive test evaluates how each method performs across different conditions and sample quantities, ensuring that the robustness of the methods is understood in a broader context. The results of these tests are shown in Table 7 and Fig. 9, which illustrate RMSE performance and statistical results of each method at various scales.

In Table 7 and Fig. 9, the iRDM-RGS method consistently outperforms the other active learning methods and random sampling across all sample sizes. The low standard deviation of iRDM-RGS also shows its strong stability. As the number of selected samples increases from 200 to 1000, the iRDM-RGS method maintains superior performance on each index. All active learning methods perform better than Random Sampling (RS). When only 200 samples are annotated, the iRDM-RGS method shows similar performance to other clustering-based methods, such as iRDM and RD, which perform well due to their high representativeness and diversity at smaller sample sizes. However, as the sample size increases, the clustering methods begin to lose their optimal properties, and the GSx method, which employs greedy selection, gradually surpasses them by selecting more diverse samples. The RD algorithm maintains diversity across multiple iterations, although it requires more computational time.

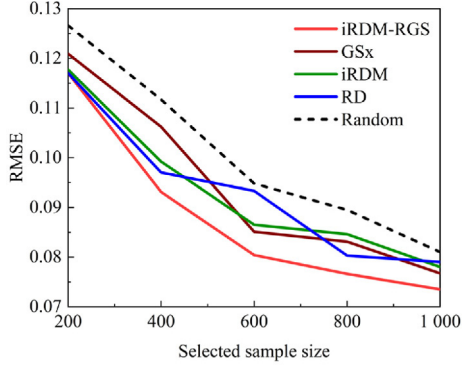
The iRDM-RGS method combines the strengths of both iRDM and greedy-based methods. Initially, iRDM selects a

Table 6 Statistical results for active learning strategies.

Strategy	RMSE		MAE	
	Mean	Std (10^{-3})	Mean	Std (10^{-3})
iRDM-RGS	0.073	2.02	0.055	1.31
GSx	0.077(+)	4.10	0.059(+)	3.85
iRDM	0.078(+)	4.09	0.059(+)	4.10
RD	0.079(+)	3.25	0.060(+)	3.90
RS	0.081(+)	2.21	0.061(+)	2.00

Table 7 Statistical results for active learning strategies under different sample sizes.

Strategy	RMSE									
	Sample size = 200		Sample size = 400		Sample size = 600		Sample size = 800		Sample size = 1000	
	Mean	Std (10^{-3})	Mean	Std (10^{-3})	Mean	Std (10^{-3})	Mean	Std (10^{-3})	Mean	Std (10^{-3})
iRDM-RGS	0.117	2.67	0.093	3.14	0.080	2.66	0.077	1.98	0.073	2.02
GSx	0.121(+)	4.32	0.106(+)	1.92	0.085(+)	2.70	0.083(+)	2.68	0.077(+)	4.10
iRDM	0.117(+)	5.87	0.099(+)	2.01	0.087(+)	2.78	0.085(+)	2.95	0.078(+)	4.09
RD	0.117(+)	5.43	0.097(+)	3.72	0.093(+)	2.94	0.080(+)	2.92	0.079(+)	3.25
RS	0.127(+)	5.95	0.106(+)	4.04	0.095(+)	2.00	0.090(+)	2.81	0.081(+)	2.21

**Fig. 9** Performance of active learning methods under different selection sample sizes.

precise subset of samples with high representativeness, and then RGS, a greedy sampling approach, is applied to the remaining samples. This combination of representativeness and diversity allows the iRDM-RGS method to maintain robust performance even as the number of annotated samples increases.

Furthermore, to demonstrate the practical impact of active learning methods, we compare the RMSE performance of selecting 800 and 1000 samples by using RS as a baseline and calculate the number of samples required for different active learning methods to achieve the same RMSE value. The results are shown in Fig. 10. While Fig. 9 illustrates the performance trends of different models across a full range of sample sizes, Fig. 10 aims to reflect the sample-saving benefits of active learning methods more intuitively. By comparing performance at 800 and 1000 samples, where models generally converge, we highlight how certain approaches can achieve comparable accuracy with fewer labeled instances. This offers a practical perspective on sample efficiency near the upper limit of model performance.

Compared to the performance of randomly selecting 800 samples, the iRDM-RGS method only requires 460 samples to achieve the same result, reducing the number of required samples by 42 %. This reduction significantly lowers the time and costs associated with obtaining labeled simulation samples, making model training more efficient. In comparison, other active learning methods require 540, 540, and 700 samples to match the performance of 800 random samples. Additionally, when compared to the random selection of 1000 samples, the iRDM-RGS method still requires only 58 % of the samples to achieve the same performance, saving 420 sam-

ple annotations in the process. This demonstrates the method's efficiency in minimizing sample labeling efforts while maintaining model performance.

To evaluate the computational cost of different active learning sampling strategies, a time comparison experiment is conducted involving all five methods. Each method is executed to select 1000 samples from the unlabeled dataset. In addition, the total simulation labeling time for these 1000 samples is measured to provide a reference for the overall SEE workflow. The results are summarized in Table 8.

As shown in Table 8, the proposed iRDM-RGS algorithm requires about 67 s to select 1000 samples. Owing to the cluster-based method requiring multiple k-means processes, which is more time-consuming than the greedy-based method, iRDM-RGS costs more time than simpler methods such as RS and GSx. Furthermore, its sampling time is significantly lower than the full cluster-based method (iRDM and RD), demonstrating improvement of computational efficiency by introducing the RGS strategy.

Moreover, although costing much time than RS and GSx, the influence of time spent on sample selection of iRDM-RGS is negligible compared to the 10 h required by the simulation-based labeling process, even with parallel computing resources. iRDM-RGS accounts for less than 0.2 % of the total data preparation time, ensuring that the additional computational cost does not hinder the overall efficiency of the SEE framework.

These results confirm that the proposed iRDM-RGS strikes a good balance between sampling quality and computational cost, making it practical for real-world UAVS effectiveness evaluation scenarios.

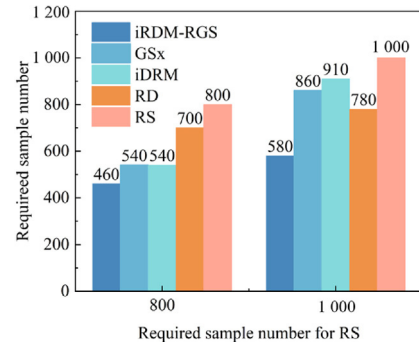
**Fig. 10** Comparison of required sample number of different methods.

Table 8 Computational time for active learning strategies.

Strategy	Sampling time (s)	Labeling time (h)
iRDM-RGS	67	
GSx	< 1	
iRDM	195	> 10
RD	8500	
RS	< 1	

6.3.3. Model structure validation and analysis

This section presents ablation studies to validate and analyze the efficacy of the two primary modules in the proposed algorithm: iRDM and RGS. These modules are compared as standalone active learning methods against the combined iRDM-RGS algorithm. Additionally, to fully evaluate the impact of the randomness introduced by RGS, a comparison is made with the non-random version of the greedy algorithm, GSx. The mean and standard deviation results are summarized in Table 9.

The iRDM-RGS method, by integrating both the iRDM and RGS modules, significantly enhances performance compared to using either module individually. As seen from the results, the RGS algorithm, which incorporates randomness through probability selection, helps the model avoid local optima, leading to better performance than the GSx algorithm, which lacks this randomness. However, when RGS is used alone, its performance is still slightly inferior to the iRDM-RGS algorithm due to its limited impact on the initial sample selection process. In contrast, the iRDM-RGS method combines the strengths of both modules. The iRDM algorithm first selects an optimal set of initial samples, ensuring a strong foundation for the model. Then, the RGS module applies random greedy sampling to these selected samples, enhancing early-stage sample selection and promoting better diversity and representativeness. This synergy of the iRDM and RGS modules results in a more efficient selection process, maximizing the value of the selected samples and boosting the overall performance of model training.

The ablation results clearly demonstrate that the iRDM-RGS approach, by focusing on high-value unsupervised samples, significantly reduces the number of required simulation annotations. This reduction in labeling efforts not only lowers simulation costs but also improves the efficiency of SEE model training for UAVS.

6.4. Utility analysis of Data-driven SEE framework

To further validate the practical utility of the Data-driven SEE Framework, this section will employ the framework on some actual instances of the established mission of UAVS.

6.4.1. Data-driven SEE for evaluation and analysis

To demonstrate the application of the Data-driven SEE model, we evaluate the effectiveness of four different UAVS SoS instances. Their SoS indicators and evaluation results are summarized in Table 10.

The four instances are evaluated using the Data-driven SEE framework. The predicted effectiveness is closely aligned with the actual simulation results, confirming the model's predictive accuracy. By including the target vehicle's SoS indicators, the framework allows for evaluation across multiple task types, enhancing its application versatility.

When comparing the instances, it is evident that higher-performing UAVS, like the second instance (No. 2), outperform lower-performance systems (No. 1) in the same mission. However, this performance improvement comes with an increased cost. Comparing No. 2 and No. 3, the UAVS setup is held constant, while the target vehicles' quantity and speed increase. The resulting drop in effectiveness demonstrates that task context alone can significantly affect system performance. Finally, from No. 3 to No. 4, task difficulty (targets' parameters) remains unchanged, but the quantity and ability of communication UAV is increased. Such enhancement facilitates the system's communication and networking ability, thereby enhancing the effectiveness of UAVS. These controlled comparisons collectively support that SoS effectiveness is jointly shaped by both system capability and task-specific contextual factors. SoS indicators of UAVS must be tailored to the specific task requirements.

6.4.2. Data-driven SEE framework for optimization

Building upon the insights gained from the previous evaluation experiment, the Data-driven SEE model can also be applied to optimize the UAVS system for improved effectiveness and reduced costs. The proposed Data-driven SEE framework not only allows for evaluating the effectiveness of UAVS systems but can also be applied to optimize their performance. Optimization typically requires running simulations, which is computationally expensive. However, by using the Data-driven model to evaluate effectiveness, we can accelerate the optimization of UAVS SoS indicators.

In this optimization problem, the effectiveness and SoS configuration cost of UAVS are set as objectives, with SoS

Table 9 Statistical results for different model structures.

Strategy	RMSE		MAE	
	Mean	Std (10^{-3})	Mean	Std (10^{-3})
iRDM-RGS	0.073	2.02	0.055	1.31
RGS	0.076(+)	3.03	0.057(+)	3.35
iRDM	0.078(+)	4.09	0.059(+)	4.10
GSx	0.077(+)	4.10	0.059(+)	3.85

Table 10 SoS indicators and effectiveness of UAVS instances.

SoS indicator		Evaluated UAVS scheme			
Type	Indicator	No. 1	No. 2	No. 3	No. 4
Detection UAV	Quantity	60	70	70	70
	Max speed	20	25	25	25
	Max acceleration	5	5	5	5
	Sensor range	200	250	250	250
Communication UAV	Quantity	30	40	40	60
	Max speed	15	25	25	30
	Max acceleration	5	5	5	6
	Sensor range	300	300	300	350
Target vehicle	Quantity	40	40	60	60
	Max speed	20	20	24	24
	Max acceleration	6	6	6	6
	Sensor range	200	200	225	225
Predicted SoS effectiveness		0.78	0.78	0.91	0.80
Actual SoS effectiveness		0.8	0.8	0.92	0.81

indicators of the UAVS serving as the decision variables. The optimization model is detailed in [Appendix B](#). The SoS configuration cost refers to the weighted cost of the UAVS system indicators, which is computed based on the eight system indicators of both detection and communication UAVS. A detailed formulation is provided in [Appendix B](#) (Eq. (B1)). Take the mission circumstance of the No.1 instance in [Table 10](#) as an example. We can optimize SoS indicators of UAVS using the multi-objective evolutionary algorithm.²⁴ Some optimized solutions are shown in [Table 11](#).

The table shows four optimized UAVS SoS indicator schemes based on the Data-driven SEE model. The effectiveness of these solutions ranges from 0.80 to 0.94, indicating strong task performance. Compared to the original SoS indicators ([Table 10](#)), the first optimized scheme has similar effectiveness but a significantly lower configuration cost. The other optimized schemes also perform better in terms of either effectiveness or cost. These optimized UAVS systems, based on the Data-driven model, can help decision-makers select the most suitable UAVS configurations for this mission.

To illustrate the simulation process, [Fig. 11](#) shows the simulation scenarios for the third optimized UAVS scheme. In [Fig. 11\(a\)](#), UAVS start from the corners and search the task area. [Fig. 11\(b\)](#) shows the mid-phase tracking process, where communication and detection UAVs collaborate to cover the entire area. [Fig. 11\(c\)](#) presents the end scenario of this mission, with successfully tracked vehicles disappearing from the map. By the task's conclusion, 95 % of the vehicles were tracked, and 27 % of the UAVs were destroyed. The final effectiveness is 0.90, close to the model's evaluation result.

These experiments demonstrate that the Data-driven SEE framework effectively evaluates the SoS effectiveness of UAVS and can be used for further optimization.

6.5. Generalization and sensitivity analysis

To verify the generalization and sensitivity of the proposed Data-driven SEE model, this section will first conduct experiments to test the performance of the SEE model on different scenarios. Then, the model's performance will also be tested

Table 11 Optimized solutions of SoS indicators.

SoS indicator		Optimized UAVS scheme			
UAV Type	Indicator	No. 1	No. 2	No. 3	No. 4
Detection UAV	Quantity	43	45	47	66
	Max speed	26.17	26.07	24.8	25.13
	Max acceleration	3.74	4.15	4.68	4.93
	Sensor range	220	220	210	240
Communication UAV	Quantity	31	35	41	60
	Max speed	7.02	6.55	5.81	10.08
	Max acceleration	2.6	2.54	4.11	9.94
	Sensor range	200	200	230	330
Objectives	Effectiveness	0.8	0.84	0.89	0.94
	Configuration cost	0.37	0.41	0.47	0.76

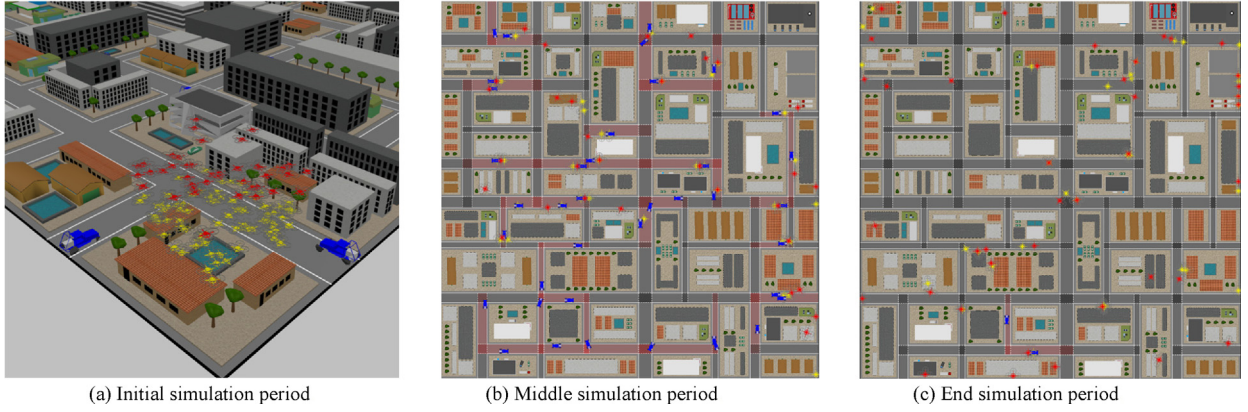


Fig. 11 Simulation process of optimized UAVS schemes.

on different combinations of weights set in Eq. (2), in order to test generalization and sensitivity of the model.

6.5.1. Generalization analysis for different scenarios

To further evaluate the generalization capability and applicability boundaries of the proposed Data-driven SEE model, we conducted transfer learning experiments across different task scenarios. The goal is to determine whether the Data-driven model, originally trained on one scenario, can be effectively adapted to a new environment with a limited number of additional samples, thus verifying its potential for fast deployment in varying operational contexts.

For the transfer learning process, the pre-trained ResNet-skew model, originally trained on the urban detection and tracking task, is fine-tuned using a small number of samples from a new target scenario. This strategy allows the model to quickly adjust to new scene characteristics without losing its previously learned knowledge, reflecting an efficient and scalable generalization mechanism.

To verify the generalization of the ResNet-skew model, three task scenarios are considered in this study:

Original Scenario (Urban detection and tracking mission): The base scenario used to train the original model, involving UAVS operations in an urban environment with detection and target tracking objectives.

Target Scenario 1 (Mountain detection and tracking mission): A new scenario featuring UAV operations in a mountainous environment. Compared to the original scenario, this task involves different terrain constraints, communication challenges, and potential sensor limitations, while maintaining a similar dual-objective (exploration and tracking) structure.

Target Scenario 2 (Urban detection mission): Another new scenario set in an urban environment but focused solely on

detection without tracking tasks. This reduces task complexity but changes mission priorities compared to the original scenario.

These two target scenarios represent both environmental and task-level variations from the original setting, providing a basis for assessing the model's adaptability.

The transfer learning process is implemented as follows: (A) The original ResNet-skew model is trained using 1000 samples from the original scenario. (B) To adapt the model to each target scenario, 200 labeled samples from the respective new scenario are added to fine-tune the model parameters. (C) Model performance is evaluated using RMSE and MAE on the separate 200 samples test set for each target scenario. Based on the above setting, the generalization performance of the ResNet-skew model is shown in Table 12. The results demonstrate that the transfer learning approach enables the SEE model to maintain good predictive performance with only a small amount of additional data from the target domain.

For scenario 1 (Mountain detection and tracking Task), the RMSE increased slightly from 0.073 to 0.082, while MAE increased from 0.055 to 0.063. These results confirm that the model effectively generalizes to new environments with similar task structures. For scenario 2 (Urban detection task), although the operating environment remains similar to the original scenario, the task structure differs (tracking is removed). As a result, the RMSE increases to 0.088 and the MAE to 0.069. While higher than scenario 1, it is still acceptable. These findings suggest that the SEE model possesses a certain degree of generalization capability and can be efficiently adapted to new but related scenarios through limited transfer learning.

While the model shows promising adaptability across moderately different scenarios, drastic changes in task structure or

Table 12 Statistical results for different scenarios.

Strategy	RMSE		MAE	
	Mean	Std (10^{-3})	Mean	Std (10^{-3})
Original	0.073	2.02	0.055	1.31
Scenario 1	0.082	2.93	0.063	2.58
Scenario 2	0.088	2.00	0.069	1.40

system configuration may require full retraining rather than fine-tuning. The model's performance depends heavily on the similarity between the source and target domains. In future work, we plan to expand the model's representational capacity by incorporating a broader range of SoS features and introducing domain adaptation techniques to enhance generalization across diverse operational environments.

6.5.2. Sensitivity analysis for weight combinations

To further verify the robustness of the SoS effectiveness calculation method in Eq. (2), a sensitivity analysis of the weight allocation between task completion rate ω_1 and disabled degree ω_2 is conducted. The weighting factor represents the relative importance assigned to these two aspects, which can be adjusted according to decision-maker preferences or mission-specific priorities. Specifically, five combinations of weights ω_1 and ω_2 are set and applied to train the ResNet-Skew model, indicating different emphasis levels on the two factors. The Data-driven SEE model is evaluated under these settings using the same 1000 training data, which contains different UAVS and target configurations. The statistical results of RMSE are recorded for comparison in Table 13.

The results show that the model's prediction performance remains stable across different weight settings, reflecting the intended effect of adjusting the weight coefficients. The model performs best when the weights for task completion rate and disabled degree are balanced (0.5:0.5), achieving the lowest RMSE and MAE. In contrast, heavily biased weights (1:0 or 0:1) lead to relatively higher errors, indicating that emphasizing only one factor reduces prediction accuracy. This suggests that a combined weighting provides richer information for the model to learn the relationship between indicators and SoS effectiveness. The chosen setting of 0.8:0.2, prioritizing task completion, also achieves low error, demonstrating a good trade-off between accuracy and mission-specific needs. This weighting is based on actual task requirements and expert recommendations, where completing the mission is considered more critical than system survivability in the evaluated UAVS scenario. Meanwhile, the low standard deviations across all settings confirm the model's stability and robustness under different weighting preferences.

This demonstrates that the SEE model is robust against moderate changes in decision-making preferences, allowing flexibility to accommodate different mission requirements without compromising predictive reliability.

6.6. Overall experimental summary

The proposed Data-driven SEE evaluation model consists of two main components: the evaluation network, ResNet-

skew, and the active learning-based sample selection strategy, iRDM-RGS. For the evaluation model, comparative experiments with recent machine learning and deep learning baselines, as well as ablation studies on the key module, have demonstrated its superior effectiveness on accuracy and in relieving the overestimation of system effectiveness in reflecting the demands. These results confirm the model's ability to capture the complex relationships among SoS indicators. For the active learning strategy, extensive validation shows that the iRDM-RGS method effectively reduces the number of required labeled samples while maintaining model performance, thereby improving annotation efficiency and reducing simulation costs.

Application experiments further verify that the proposed framework can not only evaluate the effectiveness of UAVS but also guide SoS configuration optimization. Finally, transfer learning and sensitivity analyses confirm the model's generalization ability across different task scenarios and its robustness to varying evaluation weight preferences. Overall, these comprehensive experiments validate both the effectiveness and practical applicability of the proposed method.

7. Conclusions

This paper aims to evaluate the SoS effectiveness of UAVS from a Data-driven perspective. This goal has been successfully achieved by proposing a novel Data-driven SEE framework that can efficiently evaluate the effectiveness of UAVS. In this framework, a deep learning-based evaluation network called ResNet-Skew is designed as the mainframe to perform the evaluation. In this evaluation network, SoS indicators are transformed to conform to the following ResNet structure. A skewed loss function is adopted to relieve the overestimation of the sample based on the actual application demand of the UAVS system. Besides, to reduce the high cost of simulation-based annotation, an active learning strategy iRDM-RGS is proposed to select high-value samples from large-scale unlabeled samples for training the ResNet-Skew model. The experiments show that the proposed evaluation network and active learning strategy can outperform the state-of-the-art algorithms for SEE problems. The ablation experiments' results showed that the corresponding modules of the Data-driven SEE framework are effective. Further experiments also show that the proposed framework can have potential applications in the analysis and optimization of the UAVS system. Overall, the Data-driven SEE model can quickly, precisely, and efficiently evaluate the effectiveness of UAVS.

CRedit authorship contribution statement

Wei QUAN: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Chen CHEN:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Fang DENG:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization. **Bin XIN:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization.

Table 13 Statistical results for weight combinations.

$\omega_1:\omega_2$	RMSE		MAE	
	Mean	Std (10^{-3})	Mean	Std (10^{-3})
1:0	0.082	1.75	0.057	2.14
0.8:0.2	0.073	2.02	0.055	1.31
0.5:0.5	0.065	1.22	0.051	5.89
0.2:0.8	0.071	2.16	0.052	1.88
0:1	0.079	0.87	0.061	0.40

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Simulation system of UAVS

This section provides an overview of the simulation system used to apply and validate the Data-driven model. The UAVS simulation scenario involves a collaborative detection and tracking mission^{19,24} described as follows.

In a 2 km × 2 km city environment, UAVS is commanded to detect and track ground vehicles. UAVS equipment includes drones for communication and detection. The goal for UAVS is to detect all the ground vehicles within 12 min and maintain tracking for at least 5 min without losing them.

We use the agent-based modeling method to build a simulation model of the unmanned swarm system for this scenario. In this model, each drone is considered an agent with defined motion, detection, and collaboration behaviors.

The drone model is designed for rotary-wing UAV. In 3D space, the drone's state is represented by a vector $\mathbf{Q} = [\mathbf{P}, \mathbf{V}, \mathbf{H}]$, where $\mathbf{P} = [x, y, z]$ is the position vector, $\mathbf{V} = [v_x, v_y, v_z]$ is the velocity vector, and $\mathbf{H} = [h_h, h_p]$ represents the horizontal and pitch angles. In the discrete simulation framework, the UAV updates its position based on its current velocity vector, which is controlled by speed and direction. The UAV has six motion control actions: acceleration, deceleration, left turn, right turn, ascent, and descent. These actions are combined for behaviors such as free exploration, directional tracking, and obstacle avoidance. The artificial potential field method is used for obstacle detection and avoidance. For detection, we use the truncated exponential decay disk probability sensing model⁵¹ to evaluate the UAV's detection capabilities.

Collaboration is a key feature of UAVS. The formation control and communication models are designed for two types of UAVS: communication UAV and detection UAV. The swarm is divided into groups, with each group containing a specific ratio of the two types of UAV, allowing for dynamic adjustments during the mission. The leader-follower formation control model governs the group configuration. The communication network model ensures information exchange between drones within and across groups, enabling target sharing, regional coverage, and emergency networking.²⁵

The simulation is implemented using AnyLogic software. The simulation model for collaborative detection and tracking is based on intelligent agent models, using high-frequency clock signals for continuous time simulation. The simulation control module updates the system at each clock signal inter-

val, executing scene state updates, agent actions, and system capability checks to ensure real-time behavior and task execution.

Appendix B. SoS effectiveness optimization model of UAVS

This section introduces the UAVS SoS effectiveness optimization model, further applying the Data-driven SEE framework. In terms of the optimization objective, besides SoS effectiveness, the configuration cost is included in the optimization process, which is computed as a weighted sum of the UAVS configuration indicators. It reflects the SoS-level resource requirement of a deployment plan. In this optimization problem, the decision variables are the SoS indicators of UAVS. Some indicators, like the number of drones, are discrete and usually vary within a certain integer range, while others, like maximum speed, are continuous. Thus, the decision variables are expressed as mixed variables, with discrete indicators representing quantities and sensor range and continuous ones representing speed and acceleration. Denote $\mathbf{z} = [\mathbf{x}_d, \mathbf{u}_c]$ where $\mathbf{x}_d = [x_1, x_2, \dots, x_v]$ is the vector of discrete indicators and $\mathbf{u}_c = [u_1, u_2, \dots, u_k]$ is the vector of continuous indicators.

The optimization objectives include maximizing SoS effectiveness and minimizing the configuration cost. The optimization model is formulated as follows:

$$\begin{aligned} \max_{\mathbf{z}} \quad & F_1(\mathbf{z}) = E_s = f(x_1, x_2, \dots, x_v, u_1, u_2, \dots, u_k) \\ \min_{\mathbf{z}} \quad & F_2(\mathbf{z}) = \sum_{i=1}^v c_i x_i + \sum_{j=1}^k w_j u_j \\ \text{s.t.} \quad & x_i^l \leq x_i \leq x_i^u, \forall i = 1, 2, \dots, v \\ & u_j^l \leq u_j \leq u_j^u, \forall j = 1, 2, \dots, k \\ & x_d \in \mathbf{Z}^v, u_c \in \mathbf{R}^k \end{aligned} \quad (\text{B1})$$

In this model, E_s represents effectiveness calculated by the Data-driven SEE model. c_i and w_j are the weights for calculating the configuration cost of UAVS. The second objective is referred to as "Configuration cost" in Table 11 and throughout the main text. The value constraints on x_i and u_j define their bounds, where x_i^l and u_j^l are the lower bounds of indicators, and x_i^u and u_j^u are the corresponding upper bounds. The last constraint regulates that variables are classified as discrete or continuous. The optimization seeks to maximize SoS effectiveness while minimizing the configuration cost.

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